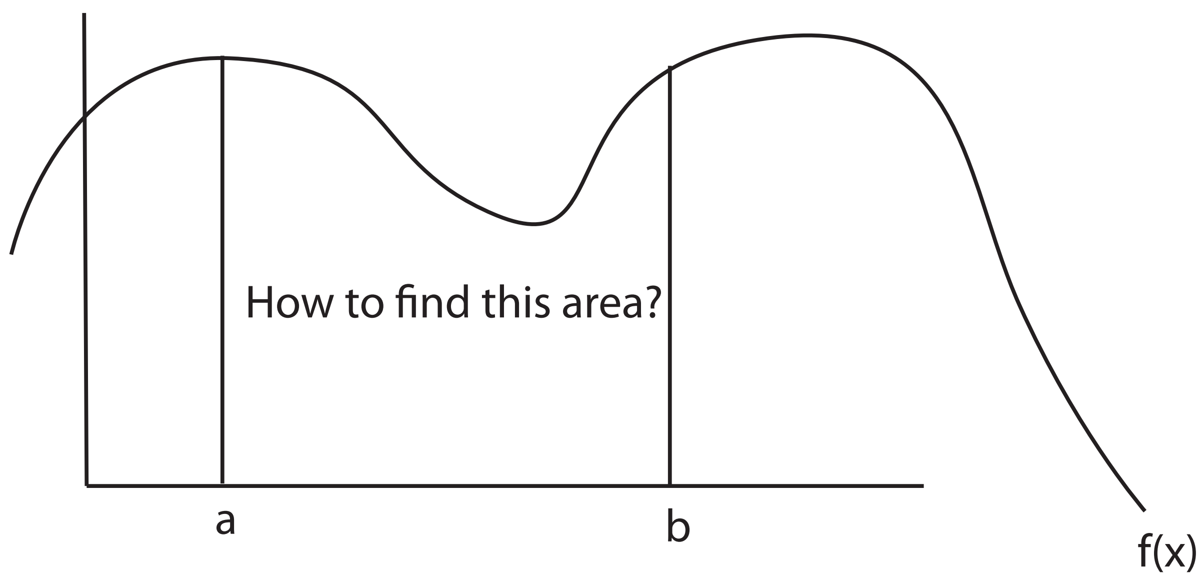
Integration

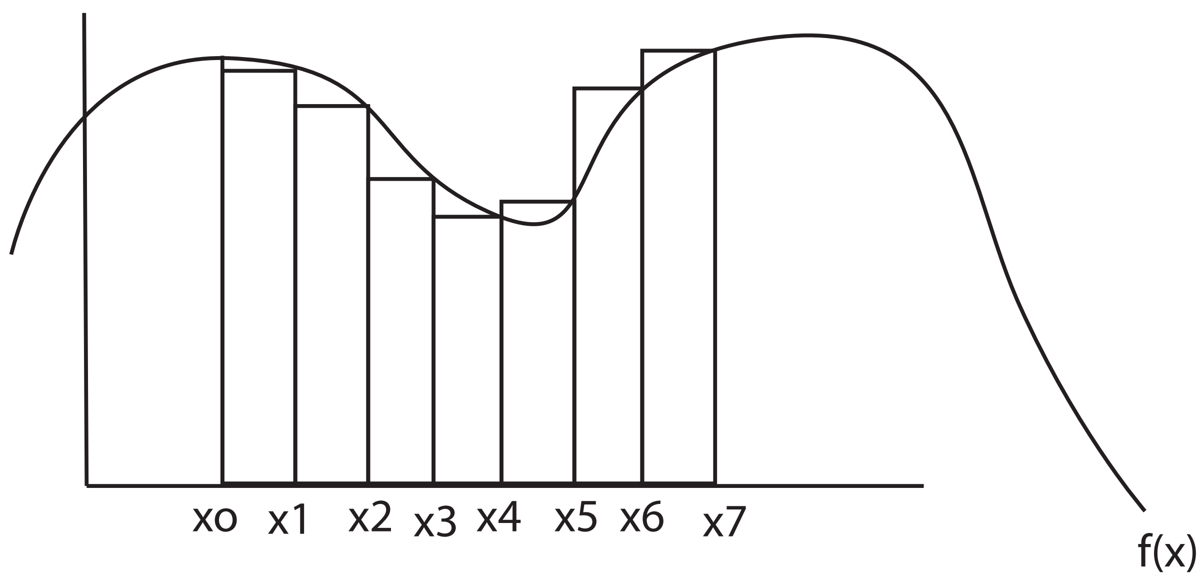
“The opposite of differentiation,” “antiderivative,” or “taking an area”

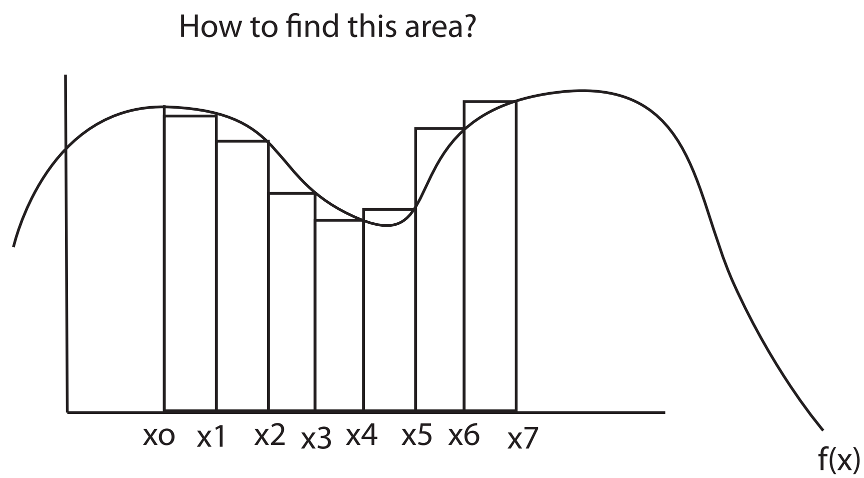


Let $a < b$ be real numbers and let f be a continuous function.

We want to find the area bounded by the curve $y = f(x)$, the vertical lines $x = a$ and $x = b$, and the x-axis. Let $a = x_o$ and $b = x_7$

How to find this area?





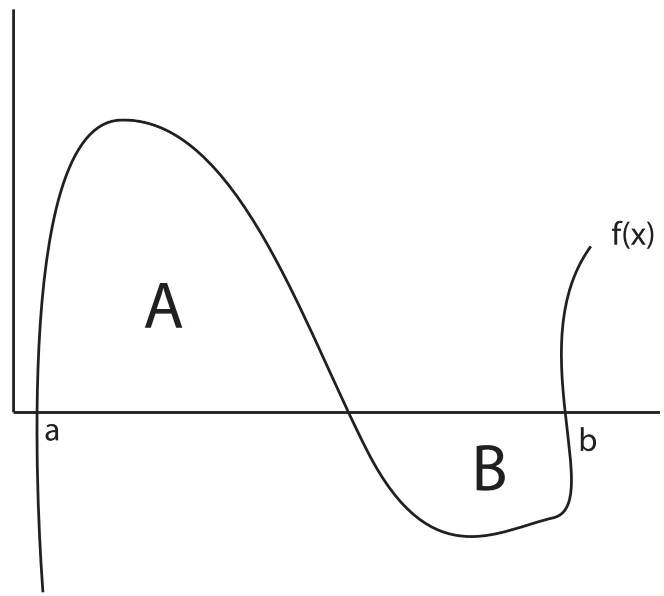
$\sum_{i=1}^{7}(x_i - x_{i-1})f(x_i)$ or letting $Dx_i = x_i - x_{i-1}$ this approximation is $\sum_{i=1}^{7} f(x_i)Dx_i$

Letting $\lim_{Dx_i} \to 0$ gives us the solution; it's written

$$\int_a^b f(x)dx$$

Areas below the $x-$axis are negative:

$$\int_a^b f(x) = \text{Area(A)-Area(B)}$$



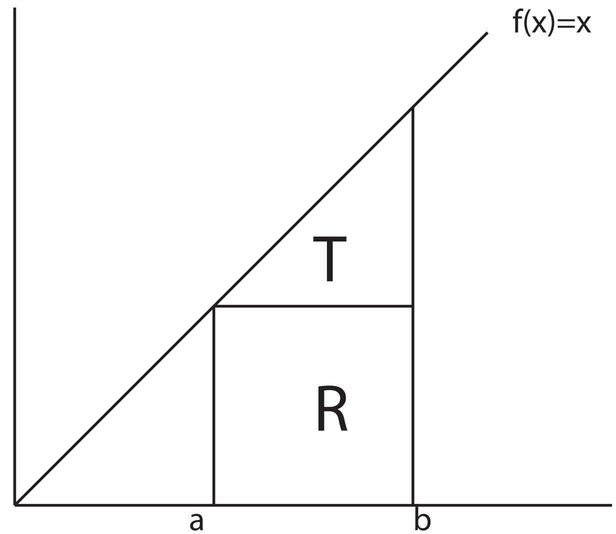
A few properties of integrals

- If k is a constant then $\int_a^b k dx = k(b-a)$
(easy to see)

- $\int_a^b x dx = \frac{1}{2}(b^2 - a^2)$

- $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$ for $a \leq c \leq b$
(and for all c in general)

$$\int_a^b x dx = \frac{1}{2}(b^2 - a^2)$$



This is $T + R$.

$$R = (b-a)(a) \text{ and } T = \frac{1}{2}(b-a)(b-a)$$

$$R + T = ab - a^2 + \frac{1}{2}b^2 - ab + \frac{1}{2}a^2 = \frac{1}{2}(b^2 - a^2)$$

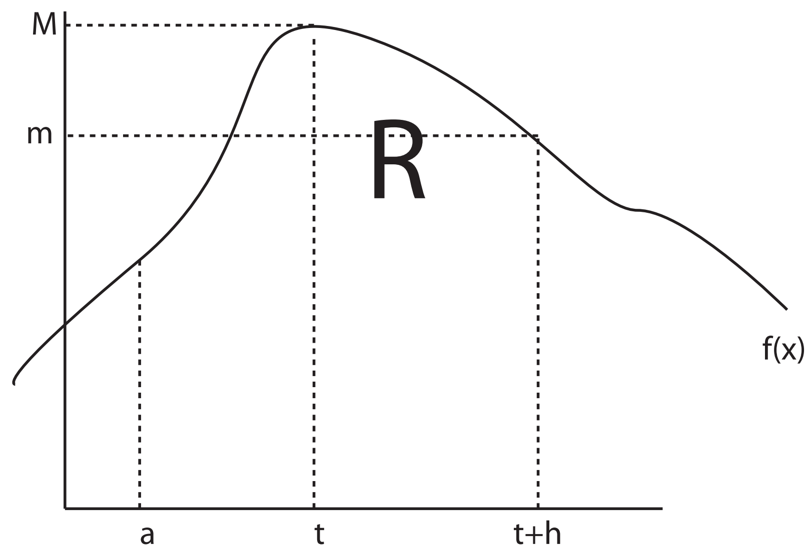
The fundamental theorem of calculus

Given a number a and a continuous function $f(x)$ we can define another function F such that

$$F(t) = \int_a^t f(x)dx \text{ for all } t$$

Then F is differentiable and $F'(t) = f(t)$ for all t

If you integrate a function and then differentiate the result, you retrieve the function that you started with $\Rightarrow$ "integration is the reverse of differentiation"



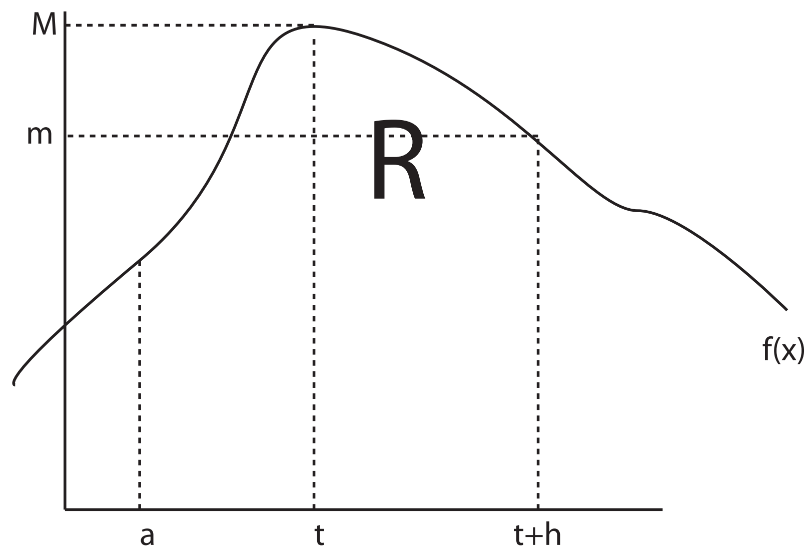
Proof: Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t. Then

$$R = \int_a^{t+h} f(x)dx - \int_a^t f(x)dx = F(t+h) - F(t)$$

We also have that

$$hm \le F(t+h) - F(t) \le hM, \text{ or}$$

$$m \le \frac{F(t+h) - F(t)}{h} \le M$$



As $h \to 0$ we have $m \to f(t) = M$. Thus,

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

implies that

$$\lim_{h \to 0} \frac{F(t+h) - F(t)}{h} = f(t)$$

or that

$$F'(t) = f(t) \qquad \square$$

Proposition: If g has a continuous derivative, then

$$\int_a^b g'(x) = g(b) - g(a)$$

Proof: Let

$$H(t) = g(t) - \int_a^t g'(x)dx$$

Then

$$H'(t) = g'(t) - g'(t) = 0 \text{ for all } t$$

Therefore H is a constant, and $H(a) = H(b)$ for all a, b.

$$H(a) = g(a) - \int_a^a g'(x)dx = g(a)$$

and since $H(a) = H(b)$, it must be that $H(b) = g(a)$ also. Therefore

$$g(a) - \int_a^a g'(x)dx = g(b) - \int_a^b g'(x)dx, \text{ or}$$

$$\int_a^b g'(x) = g(b) - g(a) \quad \square$$

Takeaway point:

To take an integral of $f(x)$ with bounds of integration (a, b) you find the function that f is the derivative of (call it $g(x)$) and then evaluate $g(b) - g(a)$.

We call g the *primitive* of f, so that

$$g'(x) = f(x) \text{ for all } x$$

A few examples that we can actually solve

Example 1: $\int_2^3 x^2 dx$

Example 2: $\int_1^2 5 - x^{-1} dx$

Example 3: $\int_{-1}^1 x dx$

Indefinite integrals

If $g(x)$ is a primitive of $f(x)$ (so that $g'(x) = f(x)$), then the set of all primitives of $f(x)$ is the set of all functions of the form $g(x) + C$, where C is any constant. C is known as the *constant of integration*

Let $f(x) = x + 7$; we know that a primitive of $f(x)$ is $g(x) = \frac{1}{2}x^2 + 7x$

It's also the case that $\tilde{g}(x) = \frac{1}{2}x^2 + 7x + 23$ is a primitive of f

Given a function f, the set of *all* its primitives is the *indefinite integral* of f, and is written

$$\int f(x)dx$$

Example: The indefinite integral $\int 12x^2 dx = 4x^3 + C$, where C is an arbitrary constant

Rule 1

$$\int x^\alpha dx = \frac{1}{\alpha+1} x^{\alpha+1} + C \text{ if } \alpha \neq -1$$

Rule 2

$$\int \frac{1}{x} dx = \ln x + C$$

Rule 3

$$\int e^x dx = e^x + C$$

Rule 1'

$$\int (x-a)^{\alpha} dx = \frac{1}{\alpha+1}(x-a)^{\alpha+1} + C \text{ if } \alpha \neq -1$$

Rule 2'

$$\int \frac{1}{(x-a)} dx = \ln(x-a) + C \text{ if } x > a$$

Rule 3'

$$\int e^{ax} dx = \frac{e^{ax}}{a} + C$$

Definite integrals

With definite integrals we do not need to worry about the constant of integration: We know that

$$\int_a^b g'(x)dx = g(b) - g(a)$$

If we used $g(x) + C$ as our primitive instead we'd get

$$\int_a^b g'(x)dx = (g(b) + C) - (g(a) + C) = g(b) - g(a)$$

How you will most likely use integration: taking expected values

Suppose that there are 10 possible states of the world that each have a different probability of arising, and that each yield a different payoff

State	a	b	c	d	e	f	g	h	i	j
Probability	.05	.1	.2	0	.1	.3	.05	0	.1	.1
Payoff	x	14	3z	2	0	0	1	2x	3	40

What is my expected payoff?

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What is my expected payoff?

(5.75+.05x+.6z)

$f(x)$ is the probability density function (pdf) of a random variable X if it satisfies the following:

- For all $a \in \mathbb{R}$, $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x) dx = 1$

The expected value of the random variable is $E[X] = \int_{-\infty}^{\infty} x f(x) dx$

The integral of a function of a random variable multiplied by the pdf gives the expected value of that function (like a weighted average)

The uniform distribution over $[a, b]$ has pdf $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$ and $f(x) = 0$ otherwise

Check that $f(x)$ is really a pdf, so that $\int_{-\infty}^{\infty} f(x)dx = 1$

Find the expected value of X when X is distributed $U[0, 1]$

What is the expected value of x^2 when X is distributed $U[0, 1]$?

What is the expected value of x^2 when X is distributed $U[2, 10]$?